

Autonomous book retrieval on TIAGo Pro

Emirates Robotics Competition 2026

Phase 1 — Simulation

Team Kanban

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Repository

github.com/4waiz/Rafeeq-library

Validated against

organiser dfl-rlab/erc_sim_2026 @ 2aa5a4a (v1.0.3-1-g2aa5a4a)

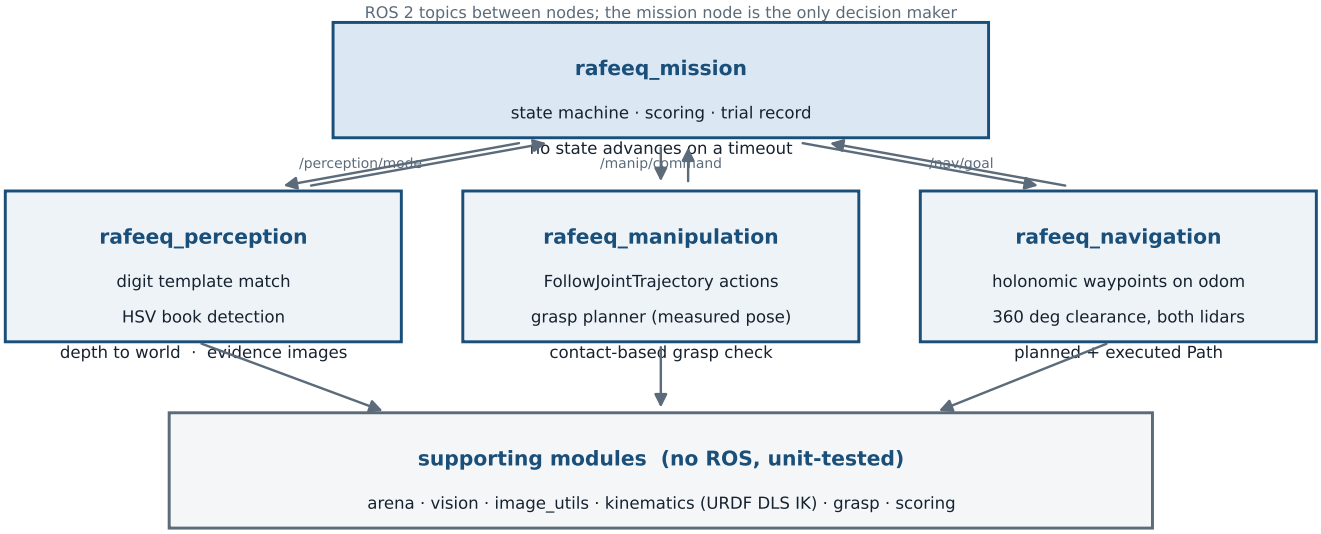
Ubuntu 22.04 · ROS 2 Humble · Gazebo Sim 8.15.0 (Harmonic)

5 recorded trials · column identified in 5/5 · book row and position in 5/5

Grasp retention is an open blocker; no points gated on it are claimed.

1 System architecture

Rafeeq is four ROS 2 nodes plus six supporting modules that carry no ROS dependency and are unit-tested offline. The mission node is the only component that decides anything: perception, navigation and manipulation each expose a narrow command-and-status contract and never talk to each other. That separation is what made the defects in this system findable at all.



Node responsibilities

Node	Responsibility	Key interfaces
rafeeq_mission	State machine, scoring, trial record	/rafeeq/*, /bin_contacts, /contacts
rafeeq_perception	Markers, books, bin, evidence images	RGB-D, CameraInfo, TF, /erc/*
rafeeq_navigation	Holonomic waypoints, clearance	/cmd_vel, /odom, /scan_*_raw
rafeeq_manipulation	Arm, torso, head, gripper	FollowJointTrajectory actions, /joint_states

Perception to action, end to end

The mission asks perception for a column; perception reads the number plates, deprojects each into the world, and publishes the physical column together with an annotated image. The mission drives there and asks for the book; perception returns the row and, critically, the measured 3D position. The mission passes that point to manipulation, which plans a grasp posture against it, and drives the base so the book is squarely in front of the arm. Nothing in this chain uses a nominal column-centre position: the organiser jitters each book by up to 0.25 m, so a nominal target misses by three times the gripper opening.

What is deliberately not used

- Nav2 is installed but never brought up by the organiser launch: no map server, no AMCL, no costmap, no map frame. Rafeeq drives odometry waypoints and publishes the intended waypoint path and the executed trajectory as nav_msgs/Path, labelled as such rather than as a Nav2 plan.
- MoveIt is installed but has nothing to plan against: the repository ships no SRDF, planning group or kinematics plugin for TIAGo Pro. kinematics.py parses the competition URDF and solves the 8-DOF chain with damped least squares.

2 Perception

Shelf number identification

The number plates are a black digit on a white square, and the exact digit textures ship inside erc_description. That makes template matching against the ground-truth textures strictly better than OCR or a learned detector here: five classes, no dataset, no training, and no failure mode that is not visible in the mask. Dark ink is segmented in HSV, each blob is tight-cropped, normalised to 48x64 and scored by intersection-over-union against every template. A digit is accepted at IoU 0.55 and must agree on three consecutive frames before it is published.

Mapping a digit to a physical column is done by measurement, not assumption: each plate is deprojected through the depth image into the world and snapped to the nearest column centre. Plates, unlike books, are not jittered - measured at exactly y = +2.0, +1.0, 0.0, -1.0, -2.0. If depth or TF is unavailable for some plates and all five are in frame, their left-to-right image order is used instead, and the fallback is logged.

Book colour, row and position

Books are unlit, fully saturated primaries presented spine-on as a 0.03 x 0.25 m rectangle, so a tight HSV band plus a 1:8 aspect test rejects almost every false positive. The row comes from two independent methods that cross-check each other: the vertical order of the four books in a column, and the measured world height snapped to the nearest row. A disagreement is reported rather than silently resolved.

The output that matters is the measured 3D position. If depth or TF is unavailable the mode fails rather than publishing a column centre - a row number alone cannot be grasped. The column filter refuses to fall back to "all books in the image": every column holds all four colours, so such a fallback would happily return a same-colour book a metre away in a neighbouring column.

[missing image: erc_images\shelf_column_20260909_100140_814.png]

Measured accuracy across the recorded trials

Quantity	Result
Trials recorded	5
Target column correctly identified	5/5
Book row identified	5/5
Book 3D position measured	5/5
Mean |offset| of book from column centre	0.145 m
Largest |offset| handled	0.254 m

Denominator is the number of recorded trials. A trial counts as identified only if the value was published from an image, never from the launch argument.

3 Navigation

The base is a mecanum holonomic drive, so x, y and yaw are independent. Rafeeq exploits that twice: it strafes along the shelf to put a jittered book squarely in front of the arm without turning away from the shelf, and it performs the entire final grasp approach as a pure base translation while the arm holds one validated posture.

Control is proportional on the odometry error, rotated into the body frame, with position tolerance 0.02 m and yaw tolerance 0.02 rad. Obstacle checking uses both 270-degree lidars binned into 36 sectors: the commanded direction is checked against the returns that actually lie in it, which a front-only test cannot do on a robot that can strafe and reverse. Stale odometry or scans stop the base rather than letting it integrate against a frozen pose. "Blocked" is reported separately from "reached": an earlier version conflated them and could believe it had arrived at a grasp pose it had never reached.

Travel time

The simulator ran at a measured real-time factor of 0.362 on the validation machine, so simulated and wall-clock seconds differ by nearly threefold. Both are recorded; the table below is simulated time, which is the machine-independent figure.

4 Manipulation

Grasping is where nearly all of the engineering went, and where the remaining blocker is. Four measured facts fix the problem almost completely.

- A book presents its spine, 0.03 m wide. Only that dimension fits between the pads, so the fingers must close along world y. This leaves the gripper roll as the single free parameter.
- The gripper body is 0.157 m long along the axis perpendicular to the fingers. With a horizontal approach that axis is vertical - and a shelf row is 0.300 m tall with a 0.250 m book already in it. The body does not fit.
- Tilting the approach rolls that body axis towards the horizontal, out of the row. Measured vertical extent of the gripper body: 0.174 m at 0 deg, 0.123 m at 45 deg, 0.060 m at 70 deg, 0.000 m at 90 deg. Whole-arm shelf intrusion reaches zero from 45 deg onwards.
- The pads themselves separate as gap = 0.0051 + 0.777 x joint, measured by transforming the fingertip collision mesh into the grasp frame. The fingertip link origins give a different and misleading curve.

The sequence

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observe -> stage 1.35 m back -> plan (roll x height x tilt) -> take posture -> base translates in   -> refine against
live base pose -> close -> verify contact -> lift -> base retreats -> verify retention
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The arm takes its posture while still 1.35 m from the shelf, where it physically cannot reach the shelf while swinging. The base then carries it in. Re-solving IK at each step of an arm-driven approach changes the arm posture inside a 0.30 m gap on every step, and the controller interpolates straight through whatever lies between two postures.

Verification, and why it is not the joint position

gz_ros2_control implements a position command interface as a JointVelocityCmd computed from position error, so the finger joint is driven kinematically and reaches whatever it is commanded to reach. Measured, commanded and achieved positions agree to five decimals whether or not a book is between the pads, and the reported effort is a constant -0.0935 N. A stall therefore cannot be used as evidence of a grasp. Rafeeq instead reads the fingertip contact sensors and requires the contact to persist through the lift and through a 0.42 m base retreat.

5 Results

run	req col/col	slot	row	offset m	grasp	deliv	coll	score	sim s
T01_seed101_col2_red	2 / red	3	1	-0.151	no	no	0	6.0	83
T02_seed202_col4_blue	4 / blue	2	4	-0.009	no	no	0	6.0	104
T03_seed303_col1_yello	1 / yellow	4	1	-0.132	no	no	0	6.0	83
T04_seed404_col5_green	5 / green	2	2	-0.254	no	no	0	6.0	121
T05_seed505_col3_red	3 / red	4	3	+0.179	no	no	1	5.5	104

Score is a LOCAL ESTIMATE from the implemented rubric, not an official result: gentle-versus-drop and collision intent are judge decisions. Times are simulated seconds; divide by the 0.362 real-time factor for wall clock.

Rates

Metric	Value	Denominator
Column identified correctly	5/5	recorded trials
Book row identified	5/5	recorded trials
Book position measured	5/5	recorded trials
Grasp verified	0/5	recorded trials
Delivered to bin	0/5	recorded trials
Collision events (total)	1	across all trials
Mean estimated score	5.9 / 19	recorded trials
Mean duration	99 sim s	recorded trials

What the failures were

- T01_seed101_col2_red: retention after lift check failed: the fingertips are no longer in contact with a book - it has been dropped
- T02_seed202_col4_blue: the book was lost during extraction: retention check check failed: the fingertips are no longer in contact with a book - it has been dropped
- T03_seed303_col1_yellow: retention after lift check failed: the fingertips are no longer in contact with a book - it has been dropped
- T04_seed404_col5_green: plan rejected: no continuous IK solution for grasp refine within 1.2 rad of the previous waypoint
- T05_seed505_col3_red: the book was lost during extraction: retention check check failed: the fingertips are no longer in contact with a book - it has been dropped

0 of 5 runs failed before the arm was used at all, at book identification: the target colour was not detected among the books the robot could see from its scan pose. 5 reached the shelf and the grasp. Because the rubric awards shelf navigation on a successful grasp and the return on a successful delivery, a failure at the grasp costs 13 of the 19 available points, not 3.

Two further trials were run and are not tabulated: they exceeded the per-trial wall-clock cap of the batch harness before reaching a terminal state, so no record was written. They are excluded rather than counted as either outcome.

6 Limitations and future work

The open blocker

Grasp retention. The evidence points at two properties of the provided environment rather than at the approach: the book collision declares max_contacts 4, so one fingertip can consume the whole budget and the opposing finger gets no contact constraint and therefore no opposing normal force; and position commands are realised as kinematic velocity setpoints, so the gripper cannot apply a clamping force at all. Increasing the squeeze from 6 mm to 16 mm of interference changed nothing. A minimal reproducer, with the exact environment version and logs, is in docs/grasp_blocker.md, and is the first thing we would raise with the organisers.

What changed during this work

Defect found	Resolution
16 states treated a timeout as success	Every transition now gates on an observation
Measured book position published, never consumed	Grasp planner consumes it; base strafes to it
Grasp poses aimed at column centres	Books jitter 0.25 m; poses solved per measurement
Top row unreachable	Wrong wrist roll; both rolls now tried
Arm drove through shelf boards	Torso set from row height; roll/tilt search
Gripper body fouled the row	Approach tilt search, 0 shelf contacts
Bin pose used the spawn height (1.3 m)	Measured settled geometry, rim at 0.950 m
Both grippers parked inside the shelf	Genuine tucks for both arms
Trajectories timed at one guessed speed	Per-joint URDF velocity limits
Evidence images excluded by .gitignore	Now committed as required evidence

Remaining work

- Resolve grasp retention, then re-run the full trial matrix and record delivery and placement outcomes.
- Replace the coarse advisory clearance model - a padded polyline through link origins - with proper swept-volume checking. It is good enough to rank IK solutions and not good enough to veto them.
- Eliminate the start-up collision in which the tucked left arm brushes the table as the base turns to face the shelf.
- Cover all 20 column/colour combinations and several randomised layouts, rather than the recorded subset.

Simulation versus physical deployment

Nothing here has run on hardware, and two of the findings in this report are specific to the simulator: the kinematic gripper and the contact budget on the book model would both behave differently on a real TIAGo Pro, where the gripper applies a real force and contact is continuous. Conversely, a physical robot brings problems this work has not had to face - odometry drift over a long run, lighting variation that would break tight HSV bands, and books that are not uniform saturated primaries. The perception approach would need re-grounding in real imagery before any of it transfers.

Longer-term intent

Rafeeq is named for the Arabic for companion, and the intent behind it is a library assistant that fetches books for people who cannot reach or navigate the stacks. None of that accessibility work is implemented: there is no voice interface, no accessibility testing and no deployment with People of Determination. It is the motivation for the project and the direction of travel, not a claim about this submission.